CAMBRIDGE INTERNATIONAL EXAMINATIONS

International General Certificate of Secondary Education

MARK SCHEME for the May/June 2014 series

0580 MATHEMATICS

0580/32 Paper 3 (Core), maximum raw mark 104

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Abbreviations

cao correct answer only
dep dependent
FT follow through after error
isw ignore subsequent working
oe or equivalent
SC Special Case
nfww not from wrong working
soi seen or implied

Qu		Answers	Mark	Part Answers
1	**(a) (i)**	5 and 9 cao	**1**	
	(ii)	4 and 9 cao	**1**	
	(iii)	8 cao	**1**	
	(iv)	2 and 5 cao	**1**	
	(b)	< = < >	**2**	**B1** for 3 correct
	(c) (i)	$(16 + 8) \div 4 - 2 = 4$	**1**	
	(ii)	$16 + 8 \div (4 - 2) = 20$	**1**	
	(d) (i)	$2 \times 2 \times 3 \times 7$	**2**	**B1** for 2, 3, 7 or 2, 2, 3, 7, or $1 \times 2 \times 2 \times 3 \times 7$
	(ii)	12	**2**	**B1** for 2, 3, 4 or 6 or $2 \times 2 \times 3$ or $2^2 \times 3$ or 4×3 or 2×6 seen as ans
	(iii)	168	**2**	**B1** for any other multiple of 168 or $2 \times 2 \times 2 \times 3 \times 7$ oe
	(e) (i)	19	**1**	any other terms must be correct
	(ii)	+4 oe	**1**	e.g. add 4
	(iii)	$4n - 1$ oe final answer	**2**	**B1** for $4n + k$, $qn - 1$ $q \neq 0$
	(iv)	accept any correct statement	**1**	

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2	**(a) (i)**	Trapezium	**1**	
	(ii)	25 200	**2**	**SCB3** for 2.52 m^2 **M1** for $\left(\frac{180+240}{2}\right) \times 120$ or $180 \times 120 + \frac{1}{2} \times 120 \times 60$ or $\left(\frac{1.8+2.4}{2}\right) \times 1.2$ or $1.8 \times 1.2 + \frac{1}{2} \times 1.2 \times 0.6$ oe
		cm^2	**1**	
	(iii)	6.3	**2**	**M1** for *their* **(a)(ii)** × 2.5 oe or figs 63
	(iv)	134 or 134.1 to 134.2	**3**	**B1** for 60 seen on diagram or used **M1** for 120^2 + (their '240 – 180')^2 or better
	(b)	correct angle bisector of angle *J* with two pairs of supporting arcs	**2**	**M1** for the correct angle bisector of angle *J* without arcs
		arc centre *H* radius 4 cm	**2**	**M1** for any arc centre *H*
		correct region shaded	**1**	dep on at least both M marks
3	**(a)**	correct mirror line	**1**	
	(b)	2	**1**	
	(c) (i)	131	**1**	
	(ii)	103	**2**	**M1** for 180 – 49 – 54 or 49 + 54 or 77 seen or fully correct method
	(d)	56	**2**	**M1** for 180 – 90 – 34 or better or indication of angle *B* = 90
	(e)	9 with supporting working	**5**	**M2** for internal angle of P =120 or **M1** for 180 – (360 ÷ 6) or (6 – 2) × 180 ÷ 6 **M1FT** for 360 – their '120' – 100 [= 140] **M1FT** for 360 ÷ (180 – their '140') if **M0** then answer of 9 scores **SC2**

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4	**(a) (i)**	2	**1**	
	(ii)	4 and a half circles	**1FT**	FT is 9 / *their* **a(i)** if *their* **a(i)** is an integer
	(b) (i)	1	**1FT**	
	(ii)	2 cao	**1**	
	(iii)	6 cao	**1**	
	(iv)	$\frac{13}{46}$ oe isw	**2**	**M1** for 13 seen or (6 + 5 + 2) / 46 or $6\frac{1}{2}$ / 23
	(c) (i)	four points correctly plotted	**2**	**M1** for 3 points correctly plotted
	(ii)	continuous ruled line of best fit	**1**	dependent on at least 9 points on graph
	(iii)	positive	**1**	
	(iv)	65 to 70	**1FT**	
	(v)	E	**1**	**FT** their continuous ruled line of best fit if positive
5	**(a) (i)**	461.7(0) cao	**1**	
	(ii)	397.06 or 397.1 or 397 or 397.062	**2FT**	**M1FT** for *their* **(a)(i)** × 0.86 oe soi
	(iii)	6880 or 6882 or 6882.(…)	**2FT**	**M1FT** for *their* **(a)(ii)** ÷ 3 soi or *their* **(a)(ii)** × 52 soi
	(iv)	84	**2**	**M1** for 140 × 3 ÷ (3 + 2)
	(b)	124 cao	**3**	**B2** for 124.3(…….) or 124.4 if **B0** then **M1** for 10 000 ÷ 80.4 **B1** for rounding their answer, if decimal, to the nearest integer
6	**(a)**	5 12	**2**	**B1**, **B1**
	(b)	9 points plotted correctly	**3FT**	**B2FT** for 7 or 8 points correctly plotted **B1FT** for 5 or 6 points correctly plotted
		correct smooth curve through all 9 correct points	**1**	
	(c)	correct ruled line	**1**	minimum length must touch *y* axis and curve
	(d)	2.7 to 2.8	**1FT**	FT their curve and ruled line

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7	**(a)**	$13p - r$ Final Answer	**2**	**B1** for either $13p$ or $-r$ in the answer or $13p - r$ spoilt
	(b)	198	**2**	**M1** for $12 \times 16 - 2 \times -3$ or **B1** for 192 or + 6 or – (–6) seen
	(c) (i)	6.4 or $6\frac{2}{5}$	**1**	
	(ii)	–3	**2**	**M1** for first correct step, i.e. $5b = 8 - 23$ or better, or $b + \frac{23}{5} = \frac{8}{5}$ or better
	(iii)	–9	**3**	**B1** for $2c - 20$ **M1FT** for correctly collecting cs on one side and numbers on the other, e.g. $5c - 2c = -7 - 20$ or better
	(d) (i)	$16x + 24$	**1**	
	(ii)	$6x\,(x - 2)$	**2**	**B1** for $x(6x - 12)$, $6(x^2 - 2x)$, $2(3x^2 - 6x)$, $3(2x^2 - 4x)$, $2x\,(3x - 6)$ or $3x(2x - 4)$
	(e) (i)	$15q^6$	**2**	**B1** for $15q^n$ (n not 0) or kq^6 (k not 0)
	(ii)	t^6	**1**	
8	**(a) (i)**	$\begin{pmatrix} 10 \\ -15 \end{pmatrix}$	**1**	
	(ii)	$\begin{pmatrix} 7 \\ -6 \end{pmatrix}$	**1**	
	(b)	$\begin{pmatrix} -4 \\ 5 \end{pmatrix}$	**1**	
	(c)	(3,1)	**1**	
9	**(a) (i)**	correct reflection at (1,–1), (3,–1) and (3,–5)	**1**	
	(ii)	correct rotation at (–1,–1), (–3,–1) and (–3,–5)	**2**	**SC1** for correct rotation any centre
	(iii)	correct translation at (–4,4), (–2,4) and (–2,8)	**2**	**B1** for one direction correct, i.e. 5 left or 3 up
	(b)	enlargement [centre] (0,1) [scale factor] 2	**1** **1** **1**	

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